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PDA Research

2019 Technology Transfer Industry Survey

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Connecting People, Science and Regulation®

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PDA Research
2019 Technology Transfer Industry Survey

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2019 Technology Transfer Industry Survey — PDA Survey Development Team

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Introduction

Technology transfers play an increasingly vital role in the regulated pharmaceutical and biopharmaceutical industries. The expanded use of contract development and manufacturing organizations (CDMO), advance of manufacturing flexibility, growth of corporate mergers and acquisitions, and increase in regulatory complexity all contribute to the need for greater efficiency in technology transfers in order to maintain continuity of drug supply. Successful execution of these transfers requires an integrated knowledge and risk management approach.

In 2014, PDA published *Technical Report No. 65: Technology Transfer,* which provides the framework for effective risk assessment and management of technology transfers ***(1)***. PDA is complementing the guidance in TR-65 with a new technical report that will provide a general overview of the steps required for successful technology transfers within the pharmaceutical and biopharmaceutical industries for active pharmaceutical ingredients (APIs) and drug products. Additionally, the new technical report will provide tools and techniques to streamline and standardize methodologies for technology transfer.

To assess current practices and learn how companies conduct technology transfers, including their technology transfer processes, knowledge and risk management systems, documentation, and business strategies, the PDA Technology Transfer Interest Group launched an industry-wide survey in early 2019. This survey assessed, across a wide range of entities, the types of technology transfers being performed, the tools and processes used in those transfers, the use of multidisciplinary teams, the potential regulatory complications, and the impact of emerging technologies.

Methodology

The survey was open to the general public for participation from January 25 to February 22, 2019. The preliminary results were aggregated and presented during the Technology Transfer Interest Group session at the *PDA Annual Meeting,* March 11-13, 2019, in San Diego, CA.

The 2019 Technology Transfer Industry Survey report is based on the views of 254 industry professionals across North America, Latin America, Europe, Asia/Pacific, Africa, and the Middle East. Survey respondents are experts involved in technology transfer activities within their current companies who possess a strong understanding of their companies' technology transfer procedures and/or activities.

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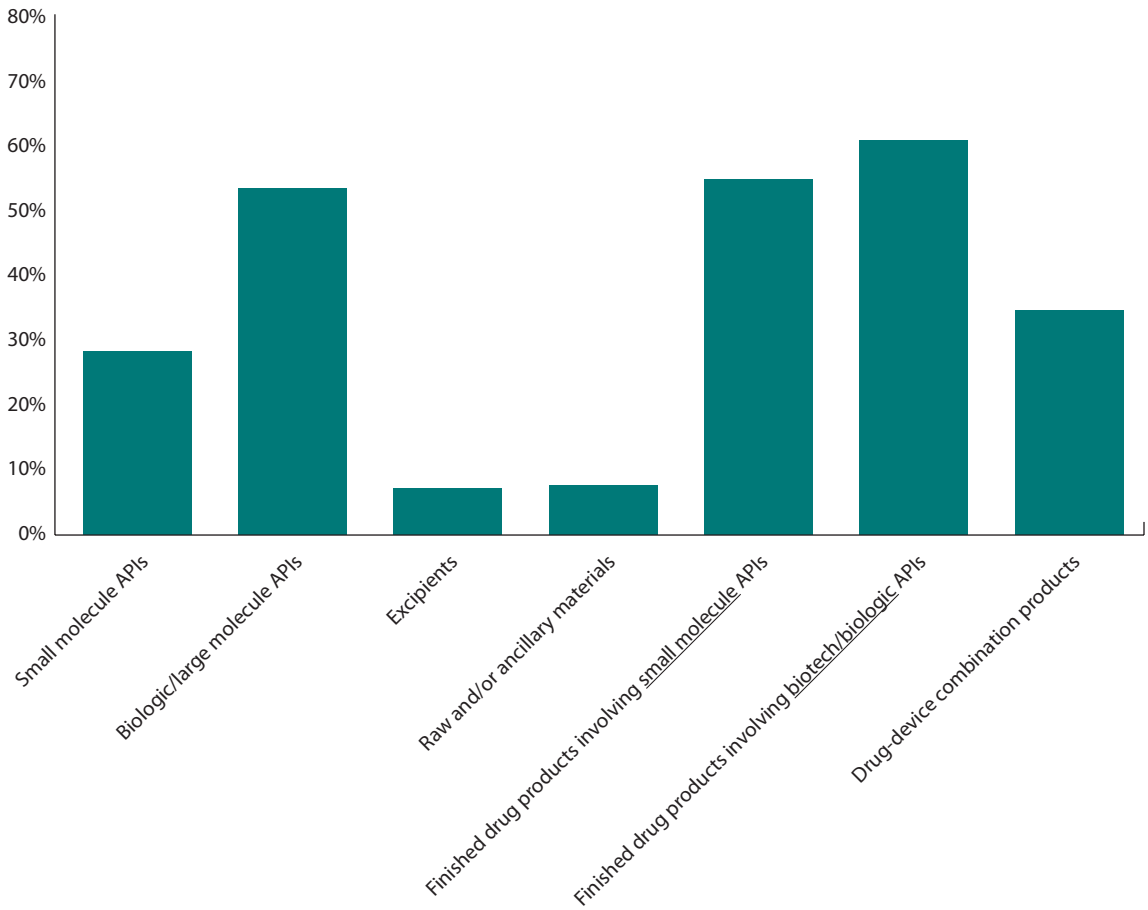
Demographics

Because technology transfers are performed across a wide range of product phases and company types, the initial survey questions queried the demographics of respondents to assess representation of the pharmaceutical and biopharmaceutical industries. The responses to the following questions demonstrate that the survey represents a broad span of industry with an equal percentage of small and large companies, multiple product types, and global reach.

1. *What types of products does your company manufacture, develop, or contract for manufacturing? (please check all that apply)*

Answer	Count	%
Small molecule APIs	71	28.0%
Biologic/large molecule APIs	134	52.8%
Excipients	18	7.1%
Raw and/or ancillary materials	19	7.5%
Finished drug products involving small molecule APIs	138	54.3%
Finished drug products involving biotech/biologic APIs	153	60.2%
Drug-device combination products	87	34.3%
	Answered	***254***
	Skipped	***0***



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2. *What is the size of your company (number of employees)?*

Answer	Count	%
Less than 100	42	16.7%
100 – 1,000	79	31.4%
1,000 – 5,000	25	9.9%
5,000 – 10, 000	18	7.1%
More than 10,000	88	34.9%
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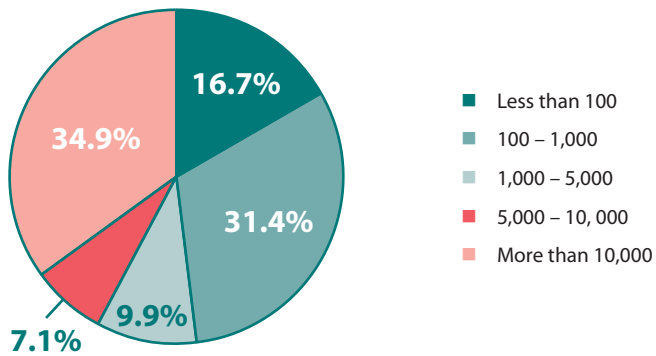
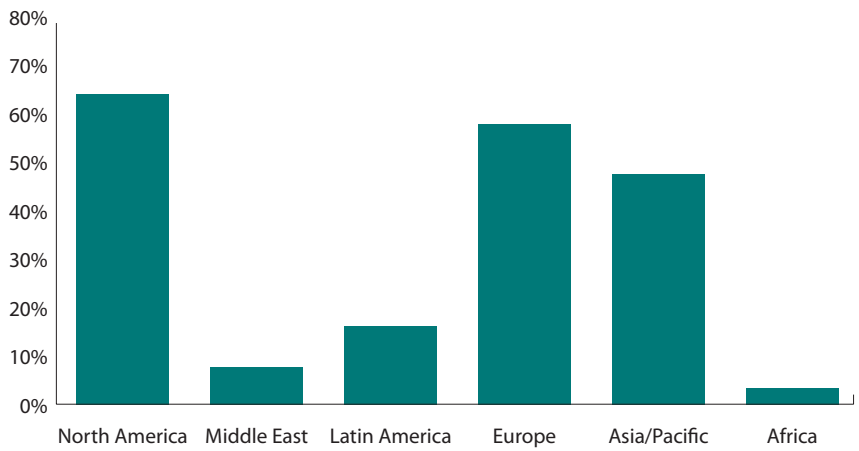
	Skipped	***2***



3. *Where are your company's internal development/manufacturing locations? (please check all that apply)*

Answer	Count	%
North America	165	65.0%
Middle East	20	7.9%
Latin America	42	16.5%
Europe	149	58.7%
Asia/Pacific	123	48.4%
Africa	9	3.5%
	Answered	***254***
	Skipped	***0***



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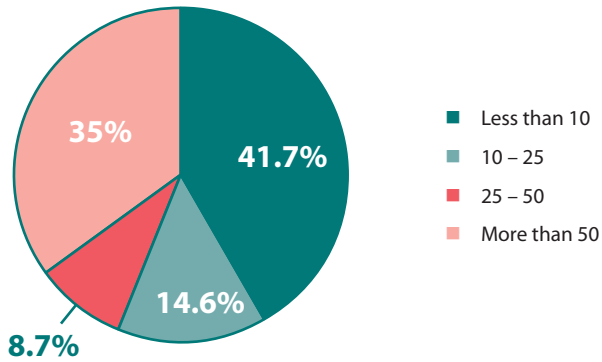
Results

Types of Technology Transfers

The survey first focused on identifying the types of technology transfers being performed. The results from these questions showed that respondents need to address knowledge transfer globally across multiple markets (Questions 4 and 5).

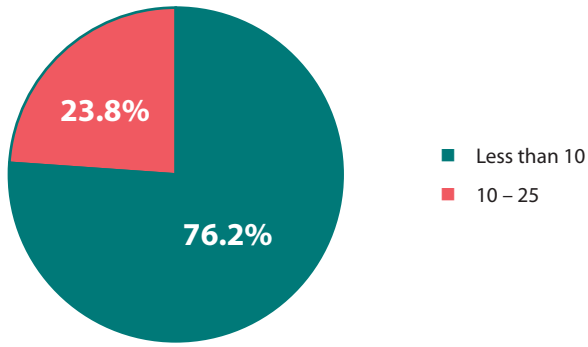
4. *In how many markets are your products licensed or used in further manufacturing?*

Answer	Count	%
Less than 10	106	41.7%
10 – 25	37	14.6%
25 – 50	22	8.7%
More than 50	89	35.0%
	Answered	***254***
	Skipped	***0***



5. *Do your technology transfers typically require knowledge transfer across multiple markets?*

Answer	Count	%
Less than 10	179	76.2%
10 – 25	56	23.8%
	Answered	***235***
	Skipped	***19***



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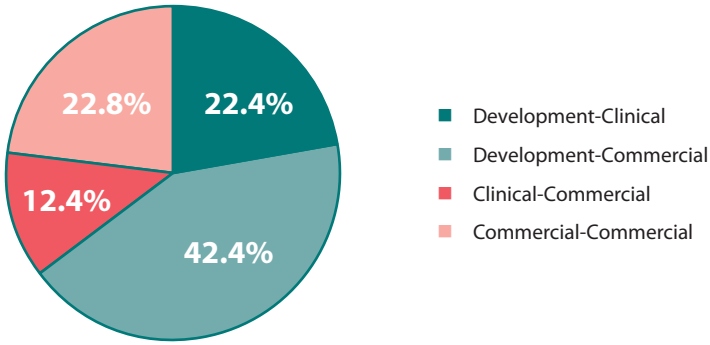
The majority of respondents said that their most common technology transfers are for new products, which are encompassed in the development-to-commercial production, development-to-clinical and clinical-to-commercial transfers; only 22.8% said their most common technology transfers are commercial-to-commercial transfers (Question 6).

6. *What is the most common type of technology transfer for your company?*

Answer	Count	%
Development-Clinical	56	22.4%
Development-Commercial	106	42.4%
Clinical-Commercial	31	12.4%
Commercial-Commercial	57	22.8%
	Answered	***252***
	Skipped	***2***



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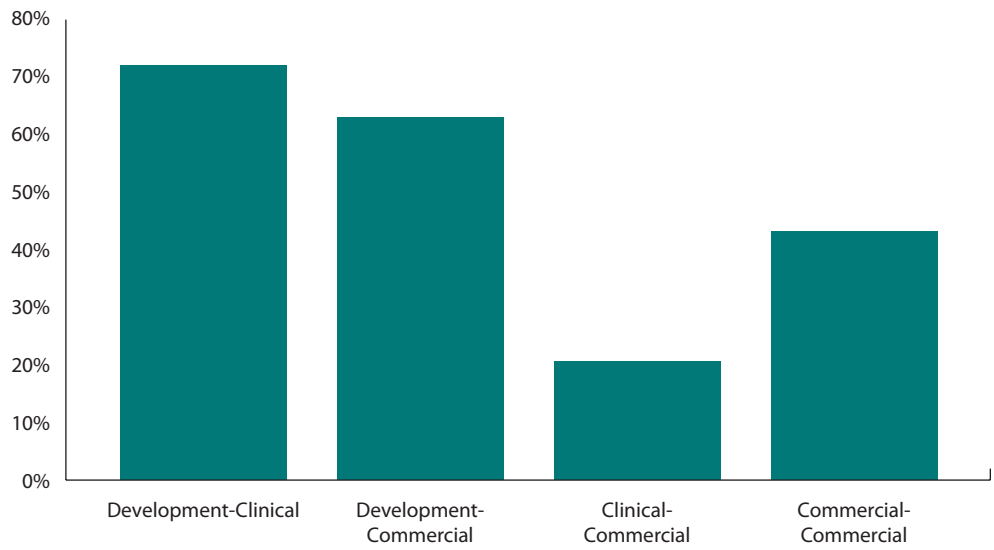
The survey showed a surprising result with 43.3% of respondents reporting product transfers from an external to an internal site (Question 7). This type of technology transfer can be a result of using a CDMO to develop a product which is then manufactured internally or a strategic decision to transfer production from a contract manufacturer to an internal site. Also, more respondents transfer technology between internal sites than outsource production through internal-to-external transfers (Question 7). This could result from mergers and acquisitions.

7. *What are the primary modes of technology transfers in your company? (please check all that apply)*

Answer	Count	%
Development-Clinical	182	72.2%
Development-Commercial	159	63.1%
Clinical-Commercial	52	20.6%
Commercial-Commercial	109	43.3%
	Answered	***252***
	Skipped	***2***



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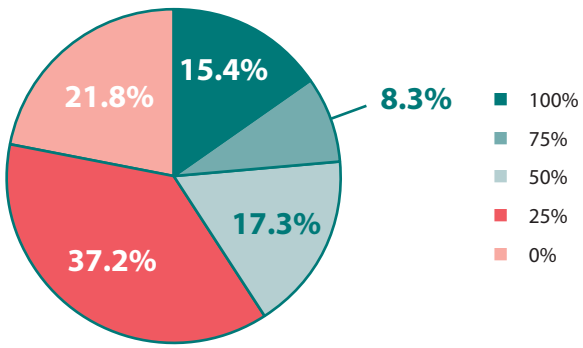
The smaller number of commercial-to-commercial product transfers mentioned above, combined with the trend to keep production internal, could indicate trends within strategic outsourcing of manufacturing that were not explored in this survey. However, in this survey, a strong 15.4% of respondents represent a new type of "virtual" company that holds the product registration but outsources all manufacturing (Question 8).

8. *Approximately what percentage of product manufacturing does your company outsource?*

Answer	Count	%
100%	24	15.4%
75%	13	8.3%
50%	27	17.3%
25%	58	37.2%
0%	34	21.8%
	Answered	***156***
	Skipped	***98***



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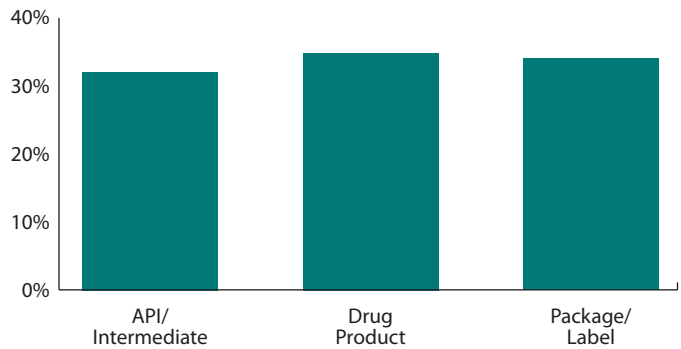
As expected, companies continue to perform some level of outsourcing for their product manufacturing. Surprisingly, respondents are outsourcing more drug product manufacturing and package/label manufacturing than API manufacturing (Question 9), which may reflect the respondents' activities, as shown in Question 1. It is not clear from the survey whether this reflects a shift toward internal manufacturing of APIs or whether respondents' companies are simply sourcing APIs from an external supplier without considering such transactions to be technology transfers.

9. *What pieces are outsourced? (please check all that apply)*

Answer	Count	%
API/Intermediate	84	33.1%
Drug Product	91	35.8%
Package/Label	89	35%
	Answered	***254***
	Skipped	***0***

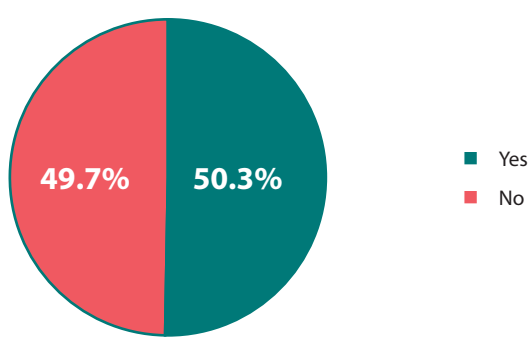


50.3% of respondents report that they are working on emerging technologies for which the existing regulatory guidance on technology transfers may not be adequate (Question 10).

For these emerging technologies, additional regulatory guidance may be necessary to successfully perform these technology transfers and achieve regulatory approval.

10. *Is your company exploring emerging technologies that may not align with current technology guidance?*

Answer	Count	%
Yes	78	50.3%
No	77	49.7%
	Answered	***155***
	Skipped	***99***



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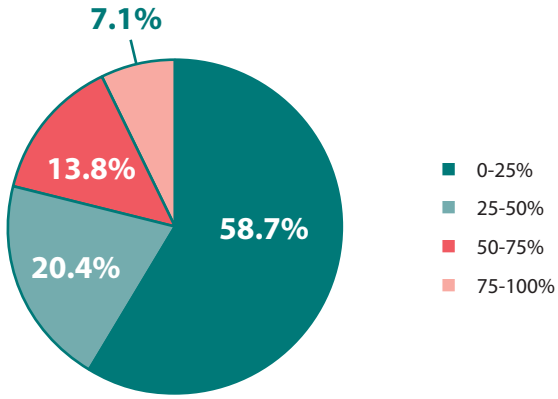
While legacy products are still being manufactured, they represent only a small percentage of the overall technology transfers being performed by respondents (Question 11).

11. *What percentage of your company's technology transfers involve legacy products approved prior to the implementation of ICH Q7, Q8, Q9, Q10, and Q11 (for which certain data may be lacking) (2, 3, 4, 5, 6)?*

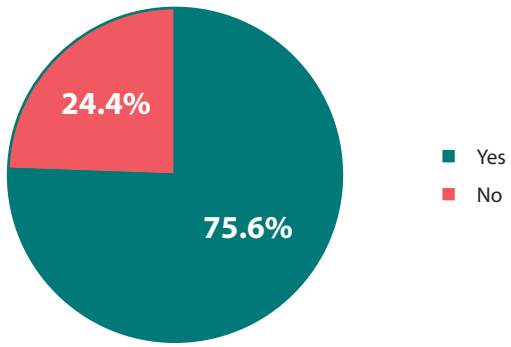
Answer	Count	%
0-25%	141	58.7%
25-50%	49	20.4%
50-75%	33	13.8%
75-100%	17	7.1%
	Answered	***240***
	Skipped	***14***



In this survey, 75.6% answered "yes" to the survey question "*Do you think a new or updated guidance is necessary to allow your company to successfully perform technology transfers?*" (Question 12).

12. *Do you think a new or updated guidance is necessary to allow your company to successfully perform technology transfers?*

Answer	Count	%
Yes	141	75.6%
No	49	24.4%
	Answered	***240***
	Skipped	***14***



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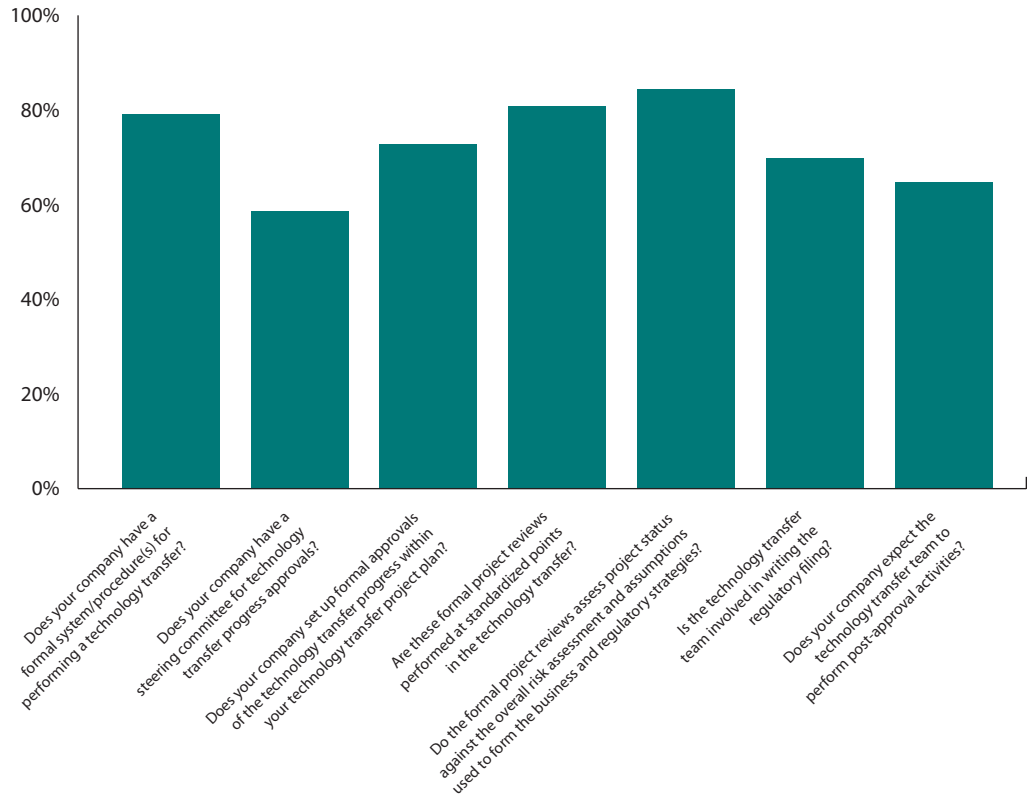
Technology Transfer Process

Using the list of questions in Table 1, the survey ascertained how responding companies perform their technology transfers. While a strong majority have a defined process for technology transfer that includes both reviews at standardized points in the process and continual assessment of project risk, 20% still do not follow a formal process as required by ICH Q10 ***(5).*** Respondents also reported that the technology transfer team is typically involved in writing the regulatory filing and then continues its engagement with the product during post-approval activities.

Answer	%
Does your company have a formal system/procedure(s) for performing a technology transfer?	79.4% Yes
Does your company have a steering committee for technology transfer progress approvals?	58.8% Yes
Does your company set up formal approvals of the technology transfer progress within your technology transfer project plan?	73.1% Yes
Are these formal project reviews performed at standardized points in the technology transfer?	81% Yes
Do the formal project reviews assess project status against the overall risk assessment and assumptions used to form the business and regulatory strategies?	84.8% Yes
Is the technology transfer team involved in writing the regulatory filing?	70.1% Yes
Does your company expect the technology transfer team to perform post-approval activities?	65% Yes

Table 1



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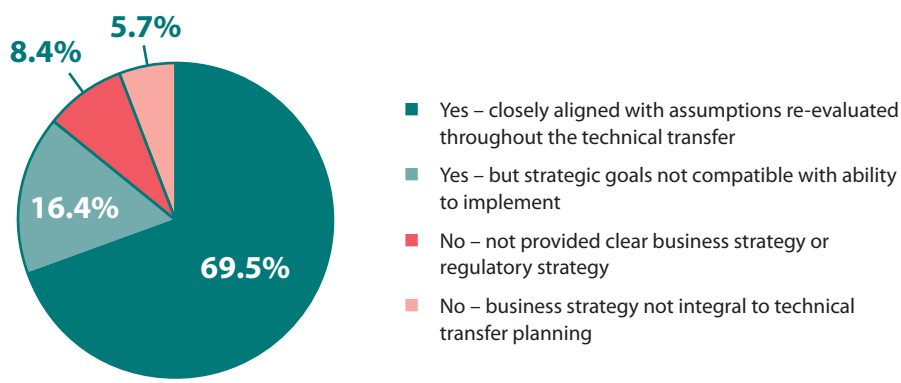
Having established that many of the technology transfer teams follow a defined process, the survey explored the effectiveness of these processes in aligning with business and regulatory goals. A key element of a successful technology transfer is achieving business goals in which the regulatory strategy plays an integral role.

The majority responded that their company does align business and regulatory strategies by reevaluating the assumptions and assessing the project's status during formal project reviews throughout the technology transfer process (Questions 13 and 14).

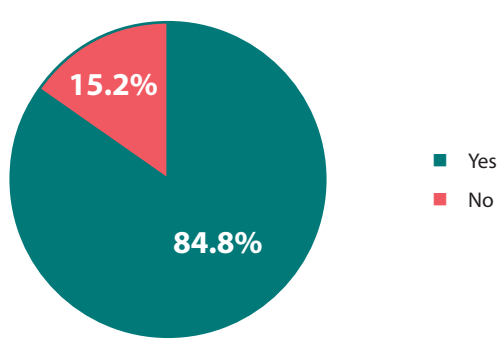
13. *In establishing project goals, does your company align business strategy with regulatory strategy?*

Answer	Count	%
Yes – closely aligned with assumptions re-evaluated throughout the technical transfer	157	69.5%
Yes – but strategic goals not compatible with ability to implement	37	16.4%
No – not provided clear business strategy or regulatory strategy	19	8.4%
No – business strategy not integral to technical transfer planning	13	5.7%
	Answered	***226***
	Skipped	***24***



14. *Do the formal project reviews assess project status against the overall risk assessment and assumptions used to form the business and regulatory strategies?*

Answer	Count	%
Yes	112	84.8%
No	20	15.2%
	Answered	***132***
	Skipped	***122***



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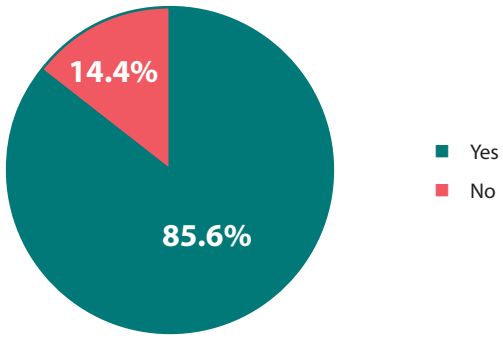
A team of people is needed to execute a technology transfer under a formal process. A large majority of respondents answered affirmatively to the question "*Does your company expect the use of a cross-functional team to implement a technology transfer?*" (Question 15).

15. *Does your company expect use of a cross-functional team to implement a technology transfer?*

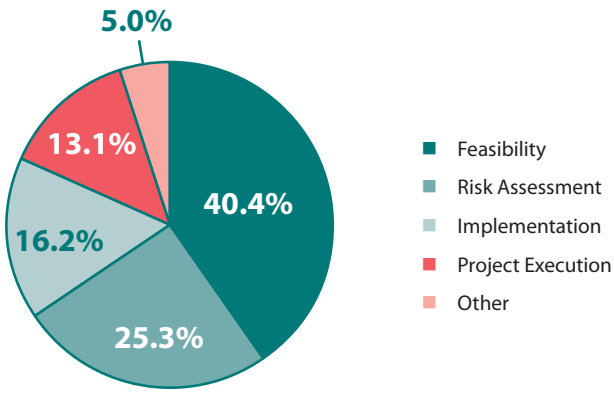
Answer	Count	%
Yes	190	85.6%
No	32	14.4%
	Answered	***222***
	Skipped	***32***



16. *At what point in the technology transfer is the cross-functional team formed? (please select the most appropriate one)*

Answer	Count	%
Feasibility	40	40.4%
Risk Assessment	25	25.3%
Implementation	16	16.2%
Project Execution	13	13.1%
Other	5	5.0%
	Answered	***99***
	Skipped	***155***

"Other" responses:
- Ad hoc – depends on project
- Also during development stage
- KO meeting with Sending Unit
- Final Formulation Selection
- Varies



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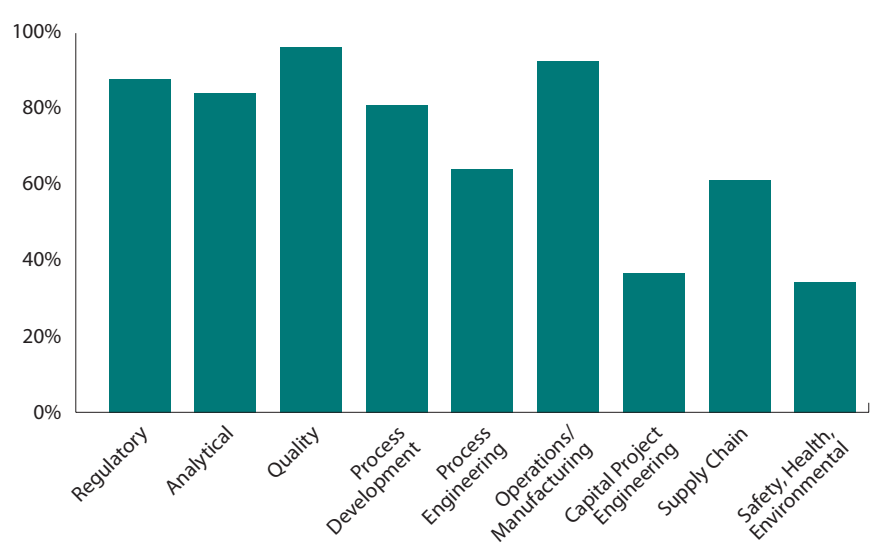
In general, companies include staff from the core functional areas (Question 17) on the technology transfer team but vary in the timing of formally involving these groups (Question 18). So, while many companies have a standardized internal technology transfer process, there appears to be limited industry standardization of the use of the cross-functional areas that are essential to achieve success.

A surprising result was how few of the technology transfer teams formally involve capital project engineering (< 40%). This may occur because the technology transfer does not require capital modifications or because the organizational structure does not lend itself to that function's involvement.

Respondents indicated that staff from the process engineering; supply chain; and safety, health, and environmental (SHE) functions are not key members of their cross-functional technology transfer teams (Questions 17 and 18). The teams that include process engineering typically do so very early in the technology transfer process, which allows for early identification and mitigation of key risks. Use of these functions will be highlighted in the new technical report, as incorporation at the right phase can facilitate early risk identification and mitigation strategies by accessing equipment and process knowledge, operational experience, and analytical strategies.

17. *What functions are incorporated as part of the cross-functional team? (please check all that apply)*

Answer	Count	%
Regulatory	115	87.8%
Analytical	110	84.0%
Quality	126	96.2%
Process Development	106	80.9%
Process Engineering	84	64.1%
Operations/Manufacturing	121	92.4%
Capital Project Engineering	48	36.6%
Supply Chain	80	61.1%
Safety, Health, Environmental	45	34.4%
	Answered	***131***
	Skipped	***123***



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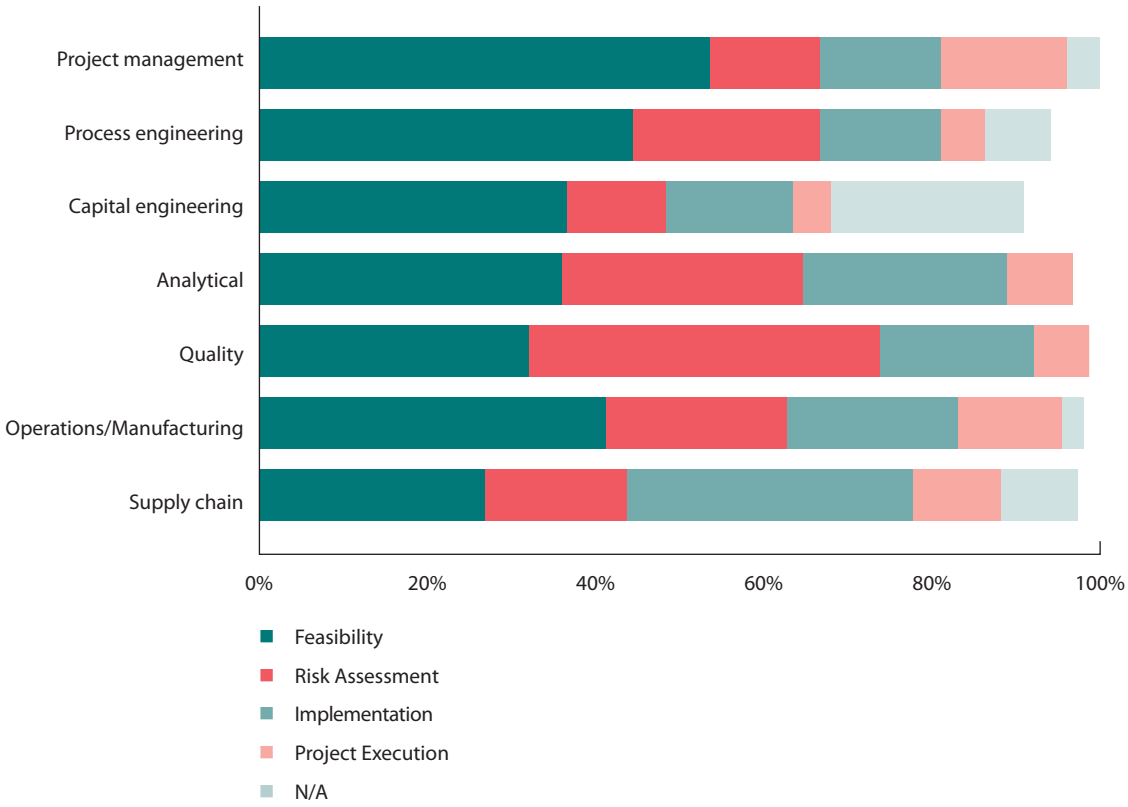
18. *At what point in a technology transfer does your company involve the following? (please select the most appropriate one)*

Answer	Feasibility	Risk Assessment	Implementation	Project Execution	N/A
Project management	82 (53.2%)	20 (13.0%)	22 (14.3%)	23 (14.9%)	6 (3.9%)
Process engineering	68 (44.2%)	34 (22.1%)	22 (14.3%)	8 (5.2%)	12 (7.8%)
Capital engineering	56 (36.4%)	18 (11.7%)	23 (14.9%)	7 (4.5%)	35 (22.7%)
Analytical	55 (35.7%)	44 (28.6%)	37 (24.0%)	12 (7.8%)	0 (0.0%)
Quality	49 (31.8%)	64 (41.6%)	28 (18.2%)	10 (6.5%)	0 (0.0%)
Operations/Manufacturing	63 (40.9%)	33 (21.4%)	31 (20.1%)	19 (12.3%)	4 (2.6%)
Supply chain	41 (26.6%)	26 (16.9%)	52 (33.8%)	16 (10.4%)	14 (9.1%)
				Answered	***154***
				Skipped	***100***



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Technology Transfer Tools

Successfully managing a technology transfer requires tools to facilitate and communicate progress of knowledge transfer for all activities, deliverables, and high-risk issues. The need for these tools is reflected in the responses to the question *"What tool/tools would you find most helpful for you for achieving successful technology transfers?"* (Question 19). While ICH guidelines provide the framework that regulators may expect to see, those guidelines are open to interpretation when it comes to implementation. The "Other" responses suggest that some companies may not have developed a robust technology transfer process. The new technical report will provide a comprehensive tool as a best practice guide to help technology transfer teams navigate and assess critical aspects of a project, determine where key roles and responsibilities should reside, and establish a communication structure.

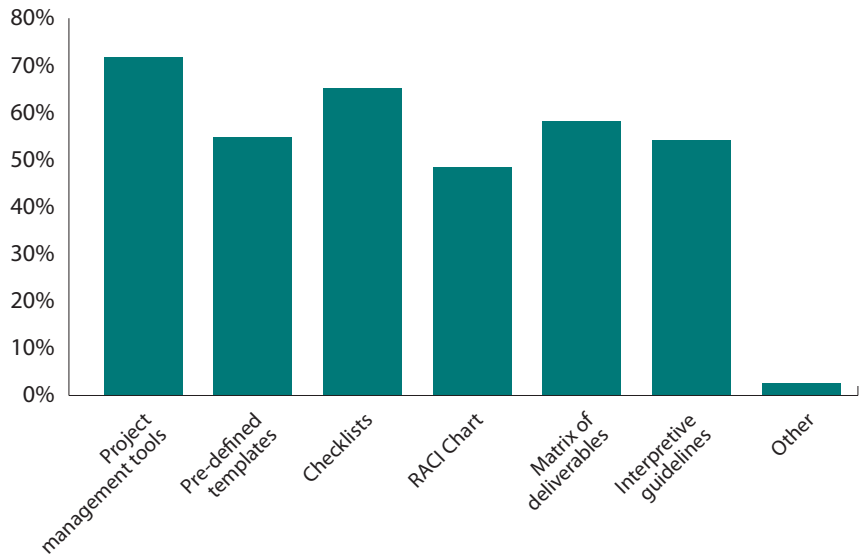
Additionally, almost half of the respondents are still using manual approvals on the documentation required for technology transfers (Question 20), which adds complexity to the overall project completion. A typical technology transfer requires a large number of documents to be written and approved; with manual approvals, document management to reduce risk of performing work prior to review and approval also becomes an extensive manual process.

19. *What tool/tools would you find most helpful for you for achieving successful technology transfers? (please select all that apply)*

Answer	Count	%
Project management tools	111	71.2%
Pre-defined templates	85	54.5%
Checklists	101	64.7%
RACI chart	75	48.1%
Matrix of deliverables	90	57.7%
Interpretive guidelines linking requirements and deliverables	84	53.8%
Other	4	2.6%
	Answered	***156***
	Skipped	***98***

"Other" responses:

- Matrix to distinguish site-specific QA requirements from product-specific requirements and gap analysis/risk assessment
- Standard manufacturing platforms
- Strategy documents
- Project plan



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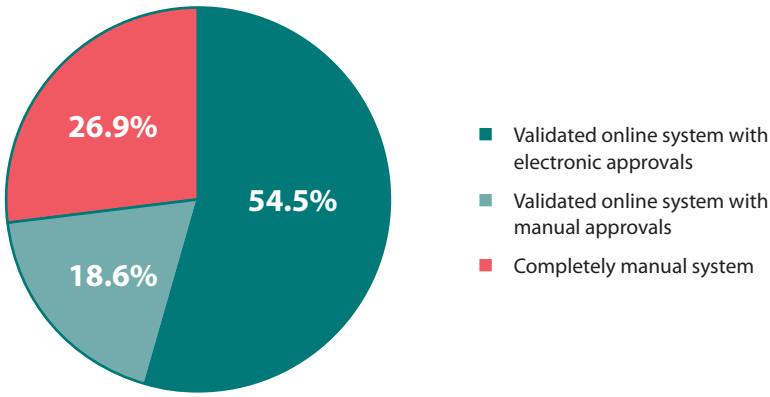
20. *What type of document management system does your company use?*

Answer	Count	%
Validated online system with electronic approvals	85	54.5%
Validated online system with manual approvals	29	18.6%
Completely manual system	42	26.9%
	Answered	***156***
	Skipped	***98***



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Challenges in Technology Transfer

Successful execution of a technology transfer requires coordination of many different elements that need to work synergistically to transfer the process and ultimately obtain regulatory approvals. According to survey respondents, the top three challenges to success are (Question 21):

- ✔ Communication/collaboration,
- ✔ Clear roles/responsibilities/boundaries, and
- ✔ Timeliness of deliverables.

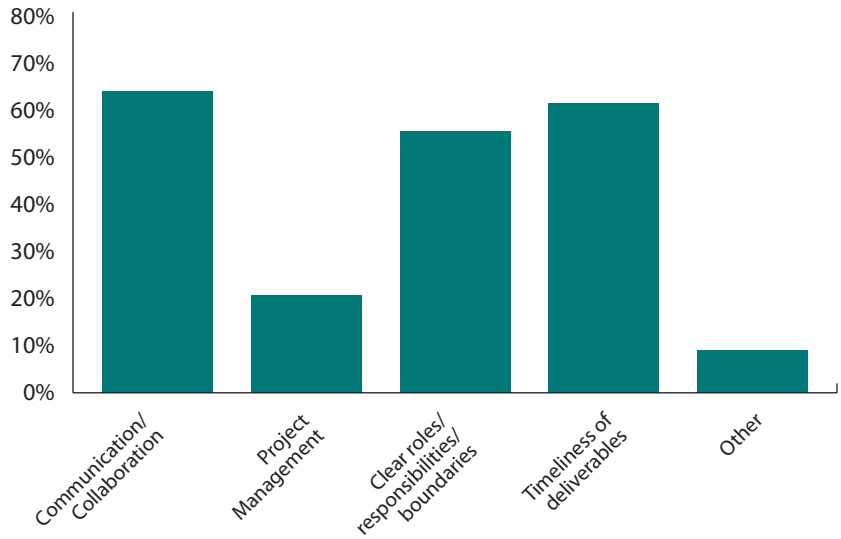
Each of these challenges could be alleviated by using a standardized framework for the technology transfer process and cross-functional team formation. Many of the additional obstacles listed in the "Other" category also indicate that some companies may not have a robust product transfer process.

21. *What do you see as your biggest obstacles for technology transfers? (please select all that apply)*

Answer	Count	%
Communication/Collaboration	99	63.5%
Project Management	32	20.5%
Clear roles/responsibilities/boundaries	86	55.1%
Timeliness of deliverables	95	60.9%
Other	14	9.0%
	Answered	***156***
	Skipped	***98***

"Other" responses:

- Change control
- Changing quality and regulatory requirements (×2)
- Internal/external prioritization
- Knowledge management & risk re-evaluation
- Lack of technical understanding of management
- Legacy processes difficult to transfer
- Long approval timelines
- QA
- Resources
- Team stability (×2)
- Technical issues not well understood
- with impact of changes
- Technical obstacles



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Conclusion

For a successful technology transfer, knowledge management needs to be integrated with a solid technology transfer process that continually assesses risk at key milestones. PDA TR 65 provides guidance on a basic technology transfer framework and risk assessment of technology transfers.

This 2019 PDA survey indicates that industry will find significant value in PDA's new technical report, which will expand the standardized framework from TR 65 and incorporate a novel grid-style tool to aid in knowledge management, boundary-setting, and cross-functional team development.

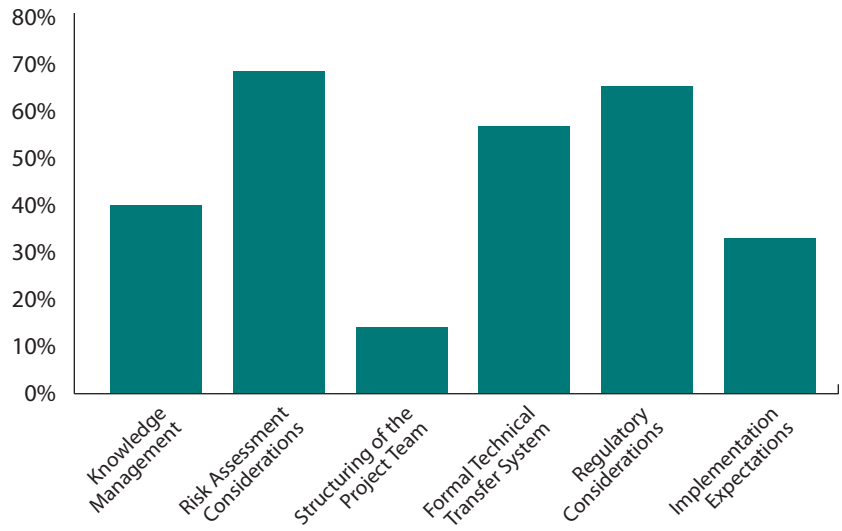
The new technical report will apply to

- Small- and large-molecule active pharmaceutical ingredients (APIs)
- Sterile and nonsterile drug products
- Packaging and labeling of all products

from development and/or clinical through commercial manufacture and should be adaptable for use with both legacy products and virtual operations.

22. *What are the top three areas for which a PDA guidance would enhance technology transfer? (please select up to three)*

Answer	Count	%
Knowledge Management	62	39.7%
Risk Assessment Considerations	106	67.9%
Structuring of the Project Team	22	14.1%
Formal Technical Transfer System	88	56.4%
Regulatory Considerations across markets, product types, technical transfer type	101	64.7%
Implementation Expectations	51	32.7%
	Answered	***156***
	Skipped	***98***



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PDA Survey: *Technology Transfer Industry*

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